PG No. 22 C_{3h} $\bar{6}$ [hexagonal]

* character table

C_{3h}	$1(1)$	$3^+_{001}(2)$	$\mathrm{m}_{001}(1)$	$-6^+_{001}(2)$
A'	1	1	1	1
E'	2	-1	2	-1
A''	1	1	-1	-1
E''	2	-1	-2	1

* polar $\leftrightarrow$ axial conversion

A' (A'') E' (E'') A'' (A') E'' (E')

* symmetric product

	A'	E'	A''	E''
A'	A'	E'	A''	E''
E'		$A'+E'$	E''	$2A''+E''$
A''			A'	E'
E''				$A'+E'$

* anti-symmetric product

A'	E'	A''	E''
$-$	A'	$-$	A'